THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 5: **Exploring Forces** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** force	Definition.
2	**(c)** smooth surfaces	Smooth.
3	**(d)** weight	Mass x gravity.
4	**(a)** newton	N.
5	**(a)** friction	It's a lubricant.
6	**(b)** at rest	Prevents motion.
7	**(d)** move a still object	Moves things.
8	**(c)** friction	Grip.
9	**(c)** friction	Surface effect.
10	**(d)** motion	Opposes relative motion.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	A push or a pull (2).	2
12	Oiling (1); Using ball bearings (1).	1+1
13	Spring balance (2).	2
14	Muscular force (1); Friction (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Changes speed (1); Changes direction (1); Changes shape (1).	1+1+1
16	Stay indoors (1); Avoid trees (1); Avoid water (1).	1+1+1
17	Pushing a door (1); Pulling a rope (1); Kicking a ball and drawing water (1).	1+1+1
	OR: Muscular (1); Friction (1); Push (1).	1+1+1
18	Attract iron (1); Have two poles (1); Like poles repel (1).	1+1+1
	OR: Force deforms it (1); Rubber is elastic (1); Regains shape after force is removed (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Starts motion (1); Stops motion (1); Changes speed (1); Changes direction (1); Changes shape (1).	1+1+1+1+1
	OR: Attracts iron (1); Poles attract and repel (1); Acts at a distance (1); Compass (1); Made into magnets (1).	1+1+1+1+1
20	Surface irregularities (1); Static (1); Sliding (1); Rolling (1); Examples (1).	1+1+1+1+1
	OR: Mass fixed (1); Weight is gravitational force (1); Moon's gravity less (1); Weight smaller (1); Mass same (1).	1+1+1+1+1

Mistakes to avoid:

- Writing that friction acts in the direction of motion; it acts **opposite** to the motion (or its tendency).
- Using kg as the unit of weight, or saying weight is constant everywhere.
- Subtracting forces that act in the same direction, or adding forces that act in opposite directions.
- Thinking friction is always harmful; it is needed for walking, brakes and gripping.
- Saying magnetic and electrostatic forces need contact, or that electrostatic force only repels.
- Concluding that an object not moving has no force on it; balanced forces also give zero motion change.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 2

Chapter 5: **Exploring Forces** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** heat	Rubbing.
2	**(d)** friction	Stop the wheel.
3	**(a)** the Earth	Downwards.
4	**(c)** friction	Rolling friction.
5	**(d)** magnetic force	Acts at a distance.
6	**(a)** iron	Magnetic materials.
7	**(a)** gravity	Earth's pull.
8	**(c)** muscular force	Needs touching.
9	**(a)** force	Force.
10	**(c)** balanced forces	Weight and normal force.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Force with which the Earth pulls objects (2).	2
12	Positive (1); Negative (1).	1+1
13	Gravitational force (1); Magnetic force (1).	1+1
14	Like poles repel (1); Unlike poles attract (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Walking (1); Brakes (1); Holding objects (1).	1+1+1
16	By rubbing with hair or wool (1); It becomes charged (1); It picks up paper bits (1).	1+1+1
17	Less friction (1); Foot slips (1); No grip (1).	1+1+1
	OR: Wear (1); Heat (1); Waste of energy (1).	1+1+1
18	Falling fruit (1); Rain falling (1); Moon orbiting the Earth (1).	1+1+1
	OR: A metal rod (1); Fixed on tall buildings (1); Connected to earth (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Walking (1); Brakes (1); Writing (1); Wear (1); Heat and noise (1).	1+1+1+1+1
	OR: Balanced: tug of war with no motion (1); Unbalanced: a pushed trolley (1); Net force zero vs not zero (1); Motion changes if unbalanced (1); Examples (1).	1+1+1+1+1
20	Rubbing causes charge (1); Positive and negative (1); Comb picks paper (1); Balloons stick to walls (1); Lightning (1).	1+1+1+1+1
	OR: Spring balance and block (1); Pull on smooth floor (1); Pull on carpet (1); Compare readings (1); Carpet shows more friction (1).	1+1+1+1+1

Mistakes to avoid:

- Writing that friction acts in the direction of motion; it acts **opposite** to the motion (or its tendency).
- Using kg as the unit of weight, or saying weight is constant everywhere.
- Subtracting forces that act in the same direction, or adding forces that act in opposite directions.
- Thinking friction is always harmful; it is needed for walking, brakes and gripping.
- Saying magnetic and electrostatic forces need contact, or that electrostatic force only repels.
- Concluding that an object not moving has no force on it; balanced forces also give zero motion change.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 3

Chapter 5: **Exploring Forces** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** attract	Opposite poles.
2	**(a)** charged	It attracts paper.
3	**(b)** metal	Conducts charge.
4	**(d)** repel	Like poles.
5	**(d)** repel	Same charges repel.
6	**(c)** electric discharge	Charge flow.
7	**(b)** a charged body touches the disc	Same charge.
8	**(b)** muscular force	By muscles.
9	**(d)** positive and negative	Two types.
10	**(b)** move a heavy object	More mass.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Gravitational force (2).	2
12	Force that opposes motion between surfaces (2).	2
13	Equal and opposite forces (1); Net force zero (1).	1+1
14	To increase friction (1); To avoid slipping (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Rough surface (1); Treads (1); Sand (1).	1+1+1
16	Mass: matter, weight: force (1); kg and N (1); Weight changes with place (1).	1+1+1
17	Compass (1); Fridge door (1); Cranes (1).	1+1+1
	OR: Oil (1); Bearings (1); Polishing (1).	1+1+1
18	Static (1); Sliding (1); Rolling (1).	1+1+1
	OR: Static acts at rest (1); Sliding acts in motion (1); Static is larger (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Parts: rod, leaves (1); Uncharged: leaves together (1); Charged body touches: leaves open (1); Same charges repel (1); Detects charge (1).	1+1+1+1+1
	OR: Ball bearings convert sliding to rolling (1); Oil forms a film (1); Surfaces do not touch directly (1); Wear reduced (1); Used in machines (1).	1+1+1+1+1
20	Charge in clouds (1); Discharge (1); Conductor (1); Stay indoors (1); Avoid water, trees (1).	1+1+1+1+1
	OR: Contact need touch (1); Non-contact act at a distance (1); Examples: friction, push (1); Examples: gravity, magnet (1); Electrostatic (1).	1+1+1+1+1

Mistakes to avoid:

- Writing that friction acts in the direction of motion; it acts **opposite** to the motion (or its tendency).
- Using kg as the unit of weight, or saying weight is constant everywhere.
- Subtracting forces that act in the same direction, or adding forces that act in opposite directions.
- Thinking friction is always harmful; it is needed for walking, brakes and gripping.
- Saying magnetic and electrostatic forces need contact, or that electrostatic force only repels.
- Concluding that an object not moving has no force on it; balanced forces also give zero motion change.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.